ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594

2. Data Isolation can help in making the data become more relevant to the objectives and the key we are looking for in the data with accuracy, while also have less error rates. We should start from a small scale first before we go into the big ones.

AI used for this answer: No

If yes, how:

External references (links):

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: Because that would remove a significant amount of data which are valuable for our tasks, thus have significant impact on our analysis that might become inaccurate. In this case, the practical solution is to simply match the missing category name with the available category number. In case that the category number is not matched or not available, we can labeled them as “Uncategorized”. In modern world, it is possible to use LLM to assists in categorize text in case that the category numbers are missing. However, this needs a strong categorization model and amount of efforts to make sure that the result of the categorization is consistent.

AI used for this answer: No

If yes, how:

External references (links):

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. Not exactly. It shows the trend of each category in overall, but cannot be compared directly in exact ratio or amount.

2. The trend random from each seed may not be the same and will be difficult to evaluate the dataset from such imbalance, which may cause the final results to be inaccurate.

AI used for this answer: No

If yes, how:

External references (links):

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. Increase.

2. This lowers the sparsity of the matrix and add more dimensions into the matrix. However, such data has low meaningful impact on the actual data mining analysis, which would cost excessive computational cost and lowers the accuracy of the data.

AI used for this answer: No

If yes, how:

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	**Number of terms kept (keep == True)**
0.0	35482
0.01	5089
0.05	1491

1. They are decreasing as the value of threshold increased.

2. There are two types of close-to-zero variance: the word that appears very rarely, and the word that appears almost every time.

3. Because they are usually the outlier or constant of data, which normally do not provide significant or clear sign of features.

AI used for this answer: No

If yes, how:

External references (links):

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

	feature	pearson_r	pearson_p
0	thanks	0.277827	2.782173e-41
1	writes	-0.233682	2.270527e-29
2	keywords	0.232965	3.395522e-29
3	looking	0.213468	1.133072e-24
4	vga	0.211099	3.755895e-24
5	video	0.209458	8.545074e-24
6	people	-0.208871	1.144445e-23
7	card	0.205532	5.934532e-23
8	hi	0.202522	2.553534e-22
9	host	0.199737	9.655856e-22

	feature	spearman_r	spearman_p
0	graphics	0.521069	2.446204e-157
1	people	-0.319349	1.115413e-54
2	files	0.316719	9.169093e-54
3	image	0.308622	5.266545e-51
4	god	-0.307345	1.408580e-50
5	3d	0.305991	3.975746e-50
6	program	0.291122	2.477587e-45
7	thanks	0.288667	1.440824e-44
8	file	0.283677	4.890639e-43
9	windows	0.276869	5.346111e-41

From the following results:

1. Spearman tends to have domain-specific terminology for graphics more than pearson in this list while word from Pearson's tends to be more generic.

2. Spearman converts values into ranks first, while Pearson tends to find correlations directly from raw values.

3. This could occurs when there are a minority of outliers that have much higher values than others in the system. While the Spearman correlation can still show reasonable

correlation values, the Pearson's correlation values will significantly drop because of the value in the outlier.

AI used for this answer: No

If yes, how:

External references (links):

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

1. Runtime is slower and the support count is less than when minSup is 9. However, the number of records when minSup of 3 is higher than when minSup is 9.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

2. Top-K FAE focuses on finding most frequent pattern, whereas the MaxFPGrowth focuses on pattern that satisfy minimum support threshold while keeping only the maximal frequent pattern.

Answer (3) — Maximal vs standard frequent patterns:

3. Because maximal frequent pattern is capable of reducing redundant results, making the output become smaller and easier to analyze while keeping similar context of the data. Data can then be represented with less patterns in the output.

AI used for this answer: No

If yes, how:

External references (links):

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. Each word in the X_counts is stored in columns, representing features and dimension of the data. Suppose we have thousands words, it means that we have thousands features, and we cannot display all thousands dimensions because each dimension represents an axis in the plot.

2. Features selection algorithms should be applied before dimensionality reduction to select the relevant features and remove the unrelated features first.

Answer: No

If yes, how:

External references (links):

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. UMAP shows distinct separated clusters, while t-SNE shows heavy overlap around the center.

2. Because the goal of PCA is to look at the overall variance of the data. PCA creates new dimension for analysis using linear approach by weighted sums the original features, which does not works well in category separation comparing to t-SNE or UMAP. t-SNE or UMAP focuses more on relationships between data which is better for categorization.

3. t-SNE's physical distance is preserved as local neighborhood; a data close to each other mostly indicates the similarity in its original high-dimensional space. Hence, t-SNE did not care about the data that is not each other's neighbor. This means that the data that looks "far" from each other might not be as far as expected.

AI used for this answer: No

If yes, how:

External references (links):

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. 4 Columns

2. An algorithm may assume that the number input has ordinal or interval meaning, when the category input is in fact nominal data.

3. One-hot binarization prevent this by changing number-labeled category (0,1,2…) to binary vector of 0 or 1, prevent ordinal or interval operations from happening.

AI used for this answer: No

If yes, how:

External references (links):
